

AHMED BEN SALAH

ICT Engineering Student | AI Engineering & Machine Learning | Cybersecurity

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PROFESSIONAL SUMMARY

ICT Engineering student at INSAT (2024–2029), Networks & Telecommunications track, with a self-directed focus on AI engineering, machine learning and cybersecurity. Builds and ships production AI systems — multi-agent RAG pipelines, LLM-driven network automation and autonomous robotics control. Two-time robotics competition podium finisher (NXP Cup, IEEE Micromouse), with certifications from Stanford/DeepLearning.AI, Harvard and Cisco. Particular interest in the intersection of AI and security.

EDUCATION

National Institute of Applied Science and Technology (INSAT) Engineering Degree in Information and Communication Technologies (ICT)

Track: Networks & Telecommunications · Self-directed focus: AI Engineering, Machine Learning, Cybersecurity
Tunis, Tunisia · 2024 – 2029 (expected)

- 5-year engineering programme covering networking, telecommunications, software engineering, AI, machine learning, data science and embedded systems.
- Coursework in network architecture, protocols and telecommunications systems — the foundation underlying the network-automation and security work below.

EXPERIENCE

Welyne · ai-commandos.com

AI & Automation Intern · April 2026 – Present · Tunis, Tunisia · On-site

- Contributed to AI Commandos, a production platform of nine specialized AI business agents, working primarily on **Hunter** (prospecting, follow-up and lead reactivation) and **Nova** (marketing and social media).
- Built LinkedIn prospecting automation for Hunter: profile scraping of high-value targets (CEOs, C-suite executives), automated multi-channel follow-up sequences and LLM-personalized outreach at scale.
- Worked on Nova's automated social campaign features — content generation, editorial calendar planning and campaign scheduling.
- Integrated LinkedIn, Facebook and Instagram publishing into the production platform, verified end to end with real posts.
- Implemented per-workspace selection of the language model and provider, and email verification with one-time codes at signup.
- Carried out a security and QA review of the deployed application, reporting issues such as authentication tokens passed in URLs and access checks that failed open.

PROJECTS

IEEE Micromouse — 3rd Place, First Micromouse Competition in Tunisia

C++ · Flood-Fill Pathfinding · PID Control · CAD · ToF Sensors · Encoders · Gyroscope · Embedded Systems
National Robotics Week, INSAT · September 2026 · Team of 5

- Placed 3rd of 7 teams with an autonomous robot that maps and solves a 16×16 maze.
- Owned the full CAD design and all of the software, including PID motion control and pathfinding.
- Implemented flood-fill maze solving, updating the distance map as walls were discovered by time-of-flight sensors.
- Built closed-loop motion on N20 motors with encoders and a gyroscope, with an on-board OLED menu for tuning parameters on the track.

NXP Cup — 3rd Place, National Autonomous Car Competition

C++ · Teensy Microcontroller · PID Control · SolidWorks · Embedded Systems · IEEE INSAT RAS

- Placed 3rd of 16 teams nationally as the youngest team in the competition, after the main Teensy board short-circuited 60 minutes before the race with no test runs available on the replacement hardware.
- Owned a hybrid software/hardware role: designed chassis components in SolidWorks for structural precision while writing the real-time PID control software for high-speed line tracking.
- Flashed firmware onto borrowed hardware with zero calibration time; the robot completed the full track and stopped 5 cm from the target cube on the second lap.

CiscoLabAI — AI-Powered Network Automation Workflow

Python · LLM Integration · Prompt Engineering · Cisco Packet Tracer · API Integration · NetDevOps

- Built an AI automation workflow that reduced Cisco Packet Tracer lab configuration from 70 minutes of manual CLI work to under 3 minutes — a 95% reduction in manual effort.
- Automated full dual-stack (IPv4 + IPv6) addressing, device hardening and advanced routing by integrating LLM-driven intent parsing with configuration-generation pipelines.
- Applied intent-based automation principles: the tool interprets architecture-level goals and generates and deploys the correct configuration sequences autonomously.
- Extending toward state-aware protocol automation (STP, inter-VLAN routing, EtherChannel).

Dalanda — Multi-Agent AI Automation System

Python · LangChain · FAISS · React · Multi-Agent AI · RAG · Prompt Engineering

- Built a multi-modal, multi-agent RAG system that ingests images, audio and documents, routing each modality to a specialized model for extraction and structuring.
- Implemented an agent verification pipeline that cross-checks generated responses against source data, reducing hallucinations without human review.
- Built API-connected workflows enabling semantic FAISS vector search with filtering, grounding LLM output in verified source data.
- Delivered a React interface with voice input, letting non-technical users drive the pipeline in natural language.

TECHNICAL SKILLS

AI Engineering	LLM Integration, Prompt Engineering, Multi-Agent Systems, RAG Pipelines, LangChain, FAISS Vector Search, Autonomous Agents, AI Workflow Automation
AI / ML	Machine Learning, Neural Networks, TensorFlow, Supervised Learning, Linear & Logistic Regression, Decision Trees, Random Forests, XGBoost, Gradient Descent, Feature Engineering, NumPy, Pandas, scikit-learn
Networking	Network Architecture, Routing & Switching, IPv4/IPv6 Dual-Stack, Protocol Analysis, CCNA Networking, NetDevOps
Cybersecurity	Network Security, Penetration Testing, Ethical Hacking, Wireshark, tcpdump, Nmap, Linux (Kali), TryHackMe, HackTheBox
Automation	Web Scraping, Workflow Automation, API Integration, Pipeline Design, Intent-Based Automation
Languages	Python, C++, C, Java, JavaScript, R, SQL, Bash/Shell Scripting
Web & APIs	Full-Stack Development, React, Node.js, REST APIs, Flask, HTML/CSS, Databases (SQL/NoSQL)
Tools & DevOps	Git, GitHub, Docker (basic), VS Code, Linux (Arch), Arduino, SolidWorks

COMPETITIVE PROGRAMMING & SECURITY PRACTICE

- **Codeforces** — active competitor ([A ZeroOne](#)); training toward TCPC and ACPC. Focus: binary search, greedy, sorting, prefix sums, two pointers.
- **TryHackMe** — [tryhackme.com/p/AZeroOne](#) — Cyber Security 101 path in progress: Linux fundamentals, Windows & Active Directory, command line, networking (Wireshark, tcpdump, Nmap).
- **PicoCTF & OverTheWire** — capture-the-flag challenges across forensics, general skills and Linux privilege exercises.

CERTIFICATIONS

Advanced Learning Algorithms	Stanford / DeepLearning.AI	Neural networks (TensorFlow), bias/variance, decision trees, XGBoost
Supervised Machine Learning: Regression and Classification	Stanford / DeepLearning.AI	Regression, classification, gradient descent, regularization
CS50: Introduction to Computer Science	Harvard / edX	Algorithms, data structures, C, Python, SQL, web development
Applied Calculus	Harvard (Verified)	Applied mathematics and quantitative reasoning
CCNA — Introduction to Networks	Cisco	Routing, switching, networking fundamentals
Certified SOLIDWORKS Associate (CSWA)	Dassault Systèmes	CAD, 3D modeling, mechanical design

LANGUAGES

Arabic (Native) · English (Professional) · French (Professional) · Spanish (Beginner)